Robustness Analysis of CNN Models for Fine-Grained Dog Breed Classification under Noisy Conditions

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***Abstract*—In this paper, we investigate the performance of convolutional neural networks (CNNs) for fine-grained dog breed classification using a large-scale dataset containing over 17,000 images across more than 100 dog breeds. Due to the high inter-class similarity among dog breeds, this task presents significant challenges for image classification models.**

Specifically, we design a complete training pipeline including data cleaning, preprocessing, and model training using two pretrained architectures: ResNet50 and EfficientNet-B0. We evaluate both models on a cleaned dataset and further examine their robustness under different noise conditions, including Gaussian noise and salt-and-pepper noise.

Experimental results show that EfficientNet-B0 achieves higher classification accuracy than ResNet50 on clean data. Furthermore, we find that both models experience significant performance degradation under Gaussian noise, while EfficientNet-B0 demonstrates stronger robustness under salt-and-pepper noise. These findings highlight the importance of model selection and robustness evaluation in real-world image classification tasks.

I. Introduction

Image classification is one of the most fundamental tasks in computer vision and has been widely applied in areas such as autonomous systems, medical imaging, and content recognition. With the advancement of deep learning, convolutional neural networks (CNNs) have achieved remarkable success in large-scale image classification tasks [1].

However, fine-grained image classification remains a challenging problem [2] . In tasks such as dog breed classification, different classes often exhibit high visual similarity, making it difficult for models to distinguish subtle differences. In addition, real-world data is often affected by noise, variations in lighting, and background complexity [3], which can significantly degrade model performance.

Most existing studies primarily focus on evaluating model accuracy on clean datasets [1], while the robustness of CNN models under noisy conditions is less explored. This creates a gap between model performance in controlled environments and real-world scenarios, where input data may contain various types of noise.

In this paper, we address this gap by conducting an empirical study on CNN-based dog breed classification using a large-scale dataset [4] containing over 17,000 images across more than 100 breeds. We design a complete training pipeline that includes data cleaning, preprocessing, and model training. Two pretrained architectures, ResNet50 [1] and EfficientNet-B0 [5], are selected for comparison.

We evaluate both models on clean validation data and further analyze their robustness under different noise conditions, including Gaussian noise and salt-and-pepper noise. As shown in Fig. 1, noise can significantly distort visual features and affect classification performance. Our results show that EfficientNet-B0 outperforms ResNet50 on clean data and demonstrates stronger robustness under certain noise conditions, particularly salt-and-pepper noise.

The main contributions of this work are summarized as follows:

- We construct a cleaned and structured dataset for fine-grained dog breed classification.
- We perform a comparative study between ResNet50 and EfficientNet-B0 on classification performance.
- We systematically evaluate model robustness under different types of noise.
- We provide insights into the impact of noise on CNN-based image classification.

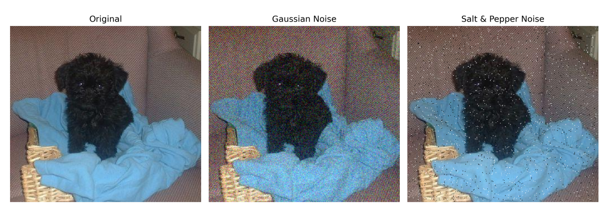


Fig. 1. Examples of clean and noisy images. Gaussian noise introduces continuous pixel perturbations, while salt-and-pepper noise introduces sparse black and white pixels.

The remainder of this paper is organized as follows. Section II reviews related work (Background and Related Work). Section III describes the methodology and experimental setup (Methods). Section IV presents the results and analysis (Experimental Results). Finally, Section **??** concludes the paper and discusses future work (Conclusions and Discussions).

II. Background And Related Work

Fine-grained image classification is usually harder than regular image classification. The reason is pretty direct. Different classes often look very similar, and the real differences are often small. Sometimes they only appear in local parts of the image. Images from the same class can still look quite different. This can happen because of pose, lighting, background, camera angle, or image quality. So the task becomes harder in two ways at once. Classes are close to each other, but samples inside one class are also not always consistent. That makes learning harder for the model. Fine-grained image classification is widely recognized as a difficult problem because of large intra-class variance and small inter-class variance, and the key visual differences are often subtle and local rather than global [8].

Dog breed classification is a very typical example of this kind of problem. Many dog breeds have similar body shape, similar face structure, and even similar fur color. If a model only looks at broad features, it can easily confuse one breed with another one. In this task, the model usually needs to notice smaller visual cues. These can include ear shape, fur texture, nose area, face proportion, and sometimes small differences around the eyes or head. These details are not always obvious. They are also easier to lose when the image is low quality or noisy. The Stanford Dogs dataset was introduced for this kind of fine-grained categorization, and it also shows that telling apart many visually similar dog breeds is not such an easy task [9]. Prior fine-grained work also points out that classification often depends on discriminative object parts or localized regions, which makes the task even more sensitive to image quality and visual detail loss [10].

Deep convolutional neural networks have been the main method for image classification for a long time. This is because they can learn image features automatically from data, instead of depending only on hand-designed features. As models got deeper, performance usually improved, but training also became harder. For fine-grained problems, CNN-based methods have been widely used because they can learn stronger representations from image regions and object parts than traditional handcrafted approaches [10]. At the same time, many studies show that fine-grained recognition often needs more than just global appearance, since highly similar categories may only differ in small local patterns [8]. This is one reason why modern deep models are so commonly used in this area.

Even though CNN models often get strong results on clean benchmark datasets, their performance under corrupted image conditions is not always studied that much. In real applications, images are often not clean. They may include Gaussian noise, salt-and-pepper noise, blur, compression artifacts, and other distortions. These changes may damage useful visual information. They can also change texture and edge details, which are often important for recognition. This matters even more in fine-grained classification. The reason is simple. Fine-grained tasks usually depend on small details more than broad object shape. So once these details are damaged, the model may still know it is looking at a dog, but it may not know which breed exactly. That difference is important.

Hendrycks and Dietterich argued that only testing models on clean data is not enough if we want to know how reliable models are in more realistic settings [11]. Their work made robustness evaluation more important in image classification research by introducing benchmarks for common corruptions and perturbations [11]. This idea is very related to our project. A model can have high accuracy on clean images, but that does not always mean it will behave well when the input becomes noisy or imperfect. In real use, people probably care about both things. They care about accuracy, but they also care whether the model stays stable when the image quality drops. In some cases, that stability may matter just as much, or even more.

So overall, image classification research has improved a lot, and CNN-based models now perform very well on many tasks. We know deep fine-grained models can learn useful local and part-based features, but their performance on fine-grained dog breed classification under poor image conditions remains under-explored [8], [10]. Specifically, few studies have conducted a true head-to-head comparison under different noise types while keeping the underlying data and evaluation settings identical. That leaves a practical gap between benchmark performance and real-world reliability. Because of this, our project compares ResNet50 and EfficientNet-B0 on clean images and also under Gaussian noise and salt-and-pepper noise. The goal is not only to see which model gets better classification accuracy. The goal is also to see which one stays more stable when important visual details become weaker, damaged, or just less clear.

III. Method

A. Research Objectives

The following research objectives are defined to guide the methodology and experimental evaluation.

O1: To evaluate the classification performance of convolutional neural network (CNN) models, specifically ResNet-50 and EfficientNet-B0, on a fine-grained dog breed dataset. *Significance:* This objective aims to assess the capability of deep learning models in distinguishing visually similar categories, which is a key challenge in fine-grained image classification.

O2: To analyze the robustness of CNN models under different noise conditions, including Gaussian noise and salt-and-pepper noise. *Significance:* In real-world scenarios, images are often affected by various types of noise due to environmental conditions or sensor limitations. Evaluating the impact of noise on model performance helps determine whether noise augmentation should be applied during training or whether preprocessing techniques, such as image denoising, are necessary to improve recognition accuracy.

O3: To compare the effectiveness of different model architectures in terms of accuracy and robustness. *Significance:* This provides insights into model selection for practical image classification tasks.

B. Data Cleaning

Data cleaning procedures were performed to improve dataset quality. Corrupted files and unsupported image formats were removed. To mitigate severe class imbalance and ensure sufficient samples for effective model learning, the dataset was filtered by removing classes containing fewer than five images. Classes with extremely limited samples are unlikely to provide meaningful feature representations and may introduce noise, thereby degrading the model's generalization performance. To identify duplicate or highly similar samples, a perceptual hashing method [6] was employed, which maps images to compact hash representations for efficient similarity comparison.

The pHash algorithm consists of the following steps:

First, the input image is converted to grayscale and resized to a fixed resolution of 32×32 pixels:

$$I' = \text{Resize}(\text{Grayscale}(I), 32 \times 32)$$

Next, a two-dimensional Discrete Cosine Transform (DCT) is applied to obtain frequency-domain coefficients:

$$D(u,v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} I'(x,y) \cos\left[\frac{(2x+1)u\pi}{2N}\right] \cos\left[\frac{(2y+1)v\pi}{2N}\right]$$

Only the top-left 8×8 low-frequency components are retained, as they capture the most significant structural information of the image. The DC component is excluded, and the mean value of the remaining coefficients is computed:

$$\mu = \frac{1}{63} \sum_{(u,v)\neq(0,0)} D(u,v)$$

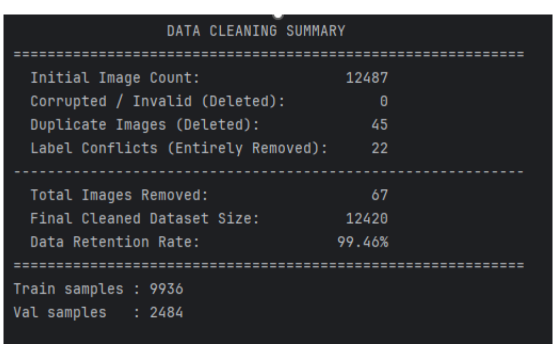
Each coefficient is then compared with the mean to generate a binary hash:

$$h(u,v) = \begin{cases} 1, & D(u,v) > \mu \\ 0, & \text{otherwise} \end{cases}$$

The resulting 8×8 binary matrix is flattened into a 64-bit hash vector. Image similarity can then be measured using the Hamming distance between hash values, where a smaller distance indicates higher similarity.

Duplicate and near-duplicate samples were removed to reduce redundancy and mitigate overfitting.

Furthermore, label conflict detection was conducted to identify instances where identical or highly similar images were assigned different labels. A total of 22 such conflicting samples were identified and removed, improving label consistency and overall data reliability.



```
DATA CLEANING SUMMARY
==========================================================
  Initial Image Count:                      12487
  Corrupted / Invalid (Deleted):                0
  Duplicate Images (Deleted):                  45
  Label Conflicts (Entirely Removed):          22
----------------------------------------------------------
  Total Images Removed:                        67
  Final Cleaned Dataset Size:               12420
  Data Retention Rate:                     99.46%
==========================================================
Train samples : 9936
Val samples   : 2484
```

Fig. 2. Data cleaning summary

C. Data Preprocessing

A statistical analysis of the image distribution across all classes was then conducted. The filtered dataset was subsequently divided into training and validation sets with a ratio of 80%:20%. All images were resized to 224×224 to ensure compatibility with standard convolutional neural network architectures. During preprocessing, images were normalized using the mean and standard deviation of the ImageNet dataset. No data augmentation was applied at this stage in order to isolate the effects of augmentation and noise in subsequent experiments.

D. Dataset Statistics

After cleaning and preprocessing, the final dataset consisted of 12,420 images across 88 classes, including 9,936 training samples and 2,484 validation samples. The training batch size was set to 32, while the validation batch size was set to 64.

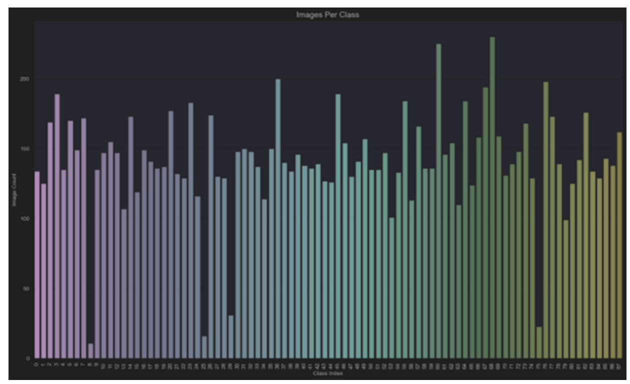


Fig. 3. Images Per Class

To better understand dataset characteristics, visualizations such as sample comparisons before and after preprocessing and class distribution histograms were generated. The analysis indicates that most classes contain between 100 and 200 images. Only a small number of classes contain fewer samples (5–50 images), while a few exceed 200 images, suggesting a relatively balanced distribution after filtering.

E. Model Training

Two widely used convolutional neural network architectures, ResNet-50 and EfficientNet-B0, were adopted for model training. Each model was trained for up to 50 epochs.

The core idea of ResNet50 is *residual learning*. Instead of directly learning a desired mapping, the network learns a residual function and adds the input back through a skip connection. This can be expressed as:

$$y = F(x, \{W_i\}) + x$$

where x is the input, $F(x, \{W_i\})$ represents the residual mapping learned by the stacked convolutional layers, and y is the output. This design helps alleviate the degradation problem and improves gradient flow, making it possible to train much deeper networks effectively.

The core idea of EfficientNet-B0 is *compound scaling*. Instead of scaling only the network depth, width, or input resolution individually, EfficientNet scales all three dimensions in a balanced way:

$$\text{depth} = \alpha^{\phi}, \quad \text{width} = \beta^{\phi}, \quad \text{resolution} = \gamma^{\phi}$$

subject to the constraint:

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$$

where ϕ is a user-specified scaling coefficient, and α, β, and γ are constants determined through optimization. This approach enables EfficientNet-B0 to achieve strong accuracy while maintaining computational efficiency.

To prevent overfitting, an early stopping strategy was implemented, where training was terminated if validation accuracy did not improve for four consecutive epochs.

The loss function used was cross-entropy loss, originally introduced in information theory [7], combined with weight decay (L2 regularization) to constrain model complexity, enhance training stability, and improve generalization performance. Cross-entropy loss is widely used for multi-class classification tasks, measuring the discrepancy between the predicted probability distribution and the ground-truth labels. It is defined as:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{1}$$

where C is the number of classes, y_i is the ground-truth label (typically one-hot encoded), and $\hat{y}_i$ is the predicted probability for class i. Minimizing this loss encourages the model to assign higher probabilities to the correct class while suppressing incorrect predictions.

F. Threats to Validity

Several factors may affect the validity of the experimental design and results. First, the experiments are conducted on a single dataset with limited types of noise, which may restrict the generalizability of the findings. In addition, each experimental setting is evaluated only once, which may introduce variability due to randomness in training and evaluation.

Furthermore, the imbalance in noise intensity makes it difficult to establish a strictly fair comparison between Gaussian and salt-and-pepper (S&P) noise. These limitations suggest that the current experimental setup may not fully reflect real-world conditions.

Future work could address these issues by incorporating more balanced noise configurations, additional datasets, and repeated experiments to improve the robustness and reliability of the study.

IV. Experimental Results

This study provides a novel perspective by combining fine-grained image classification with robustness analysis under noisy conditions. While convolutional neural networks such as ResNet and EfficientNet have been widely studied for image classification, their performance under different types of noise has not been extensively compared in the context of fine-grained dog breed classification.

In particular, this work systematically evaluates the impact of Gaussian and salt-and-pepper (S&P) noise on model performance and investigates how different architectures respond to noise-induced degradation. This analysis offers practical insights into model robustness and supports informed decisions regarding data augmentation and preprocessing strategies in real-world applications.

To comprehensively evaluate model performance from multiple perspectives, a set of evaluation metrics including accuracy, precision, recall, and F1-score is adopted. These metrics enable a detailed analysis of both classification accuracy and the model's behavior under different conditions.

Precision evaluates the correctness of positive predictions, indicating how often the model avoids misclassification among visually similar classes. Recall measures the model's ability to correctly identify instances of each class, reflecting its sensitivity to all categories. The F1-score, as the harmonic mean of precision and recall, provides a balanced evaluation of classification performance, particularly when both false positives and false negatives are important.

Macro and weighted averages are reported to assess performance across all classes. The macro average treats each class equally, highlighting the model's consistency across different categories, while the weighted average accounts for the number of samples per class, reflecting overall dataset-level performance.

The performance of ResNet-50 and EfficientNet-B0 on the validation set is summarized in figure 4. EfficientNet-B0 consistently outperforms ResNet-50 across all evaluation metrics.

Specifically, EfficientNet-B0 achieves an overall accuracy of 97.2%, compared to 96.1% for ResNet-50. In terms of macro-averaged metrics, EfficientNet-B0 attains a precision, recall, and F1-score of 0.971, while ResNet-50 achieves 0.963, 0.961, and 0.961, respectively. Similarly, the weighted averages also indicate improved performance for EfficientNet-B0.

These results suggest that EfficientNet-B0 provides better feature representation and generalization capability, leading to more accurate and balanced classification performance across different classes.

ResNet50

	precision	recall	f1-score	support
accuracy			0.961	2484
macro avg	0.963	0.961	0.961	2484
weighted avg	0.962	0.961	0.961	2484

EfficientNetB0

	precision	recall	f1-score	support
accuracy			0.972	2484
macro avg	0.971	0.971	0.971	2484
weighted avg	0.973	0.972	0.972	2484

Fig. 4. Classification performance comparison

The confusion matrices of EfficientNet-B0 and ResNet50 both show strong diagonal dominance, indicating high overall classification accuracy. However, EfficientNet-B0 exhibits a cleaner diagonal with fewer off-diagonal errors, suggesting more accurate and consistent predictions across classes.

In contrast, ResNet50 shows slightly more scattered misclassifications, particularly among visually similar categories, which is expected in fine-grained classification tasks. These results indicate that EfficientNet-B0 learns more discriminative feature representations and is better suited for distinguishing subtle differences between dog breeds.

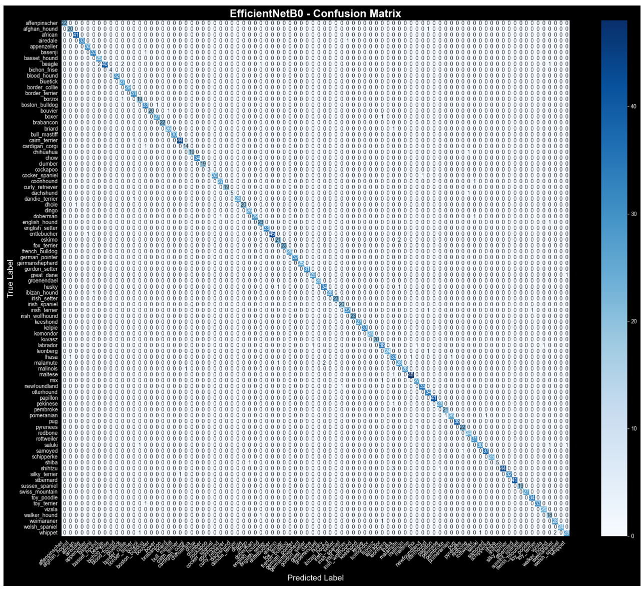


Fig. 5. EfficientNet-B0

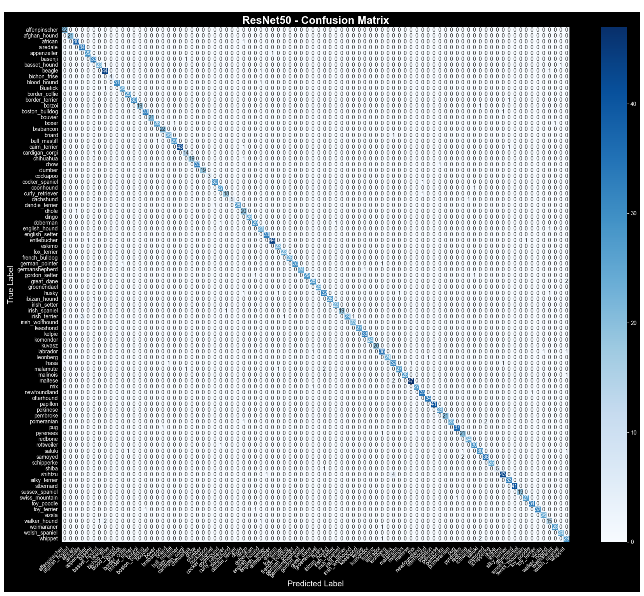


Fig. 6. ResNet-50

To evaluate model robustness under noisy conditions, an augmented version of the validation set was constructed by introducing Gaussian noise, salt-and-pepper noise.

The performance of the model on this modified validation set was compared with the original validation set to assess the robustness. Experimental results show that the introduction of Gaussian noise led to a significant accuracy drop of approximately 40% for both models. When salt-and-pepper noise was applied, the accuracy decreased by 14% and 1% for the two models, respectively.

Model	Validation Loss	Accuracy (%)	Top-3 Accuracy (%)
ResNet50 (Clean)	0.2170	96.10	98.55
EfficientNetB0 (Clean)	0.1598	97.18	99.19
ResNet50 (Gaussian)	2.0099	56.76	75.04
EfficientNetB0 (Gaussian)	1.7163	59.22	77.82
ResNet50 (S&P)	0.8313	82.69	92.55
EfficientNetB0 (S&P)	0.2307	95.93	98.55

Fig. 7. Robustness Evaluation Results

The performance of ResNet-50 and EfficientNet-B0 under Gaussian noise and salt-and-pepper noise is summarized in figure 8 and 9.

Under Gaussian noise, both models experience significant degradation in performance. ResNet-50 achieves an accuracy of 56.8%, while EfficientNet-B0 performs slightly better with an accuracy of 59.2%. A similar trend is observed in the F1-score, where EfficientNet-B0 (0.597) outperforms ResNet-50 (0.580). These results indicate that Gaussian noise severely impacts model performance and that EfficientNet-B0 demonstrates relatively stronger robustness under this condition.

In contrast, under salt-and-pepper noise, both models maintain considerably higher performance. ResNet-50 achieves an accuracy of 82.7%, whereas EfficientNet-B0 achieves a significantly higher accuracy of 95.9%. The F1-score also shows a substantial gap, with EfficientNet-B0 reaching 0.959 compared to 0.830 for ResNet-50. This suggests that EfficientNet-B0 is more resilient to impulse noise and is able to preserve discriminative features more effectively.

```
======================================================================
CLASSIFICATION REPORT — Gaussian Noise
======================================================================
ResNet50 (Gaussian)
                  precision    recall  f1-score   support

        accuracy                         0.568      2484
       macro avg      0.712     0.570     0.583      2484
    weighted avg      0.713     0.568     0.580      2484

EfficientNetB0 (Gaussian)
                  precision    recall  f1-score   support

        accuracy                         0.592      2484
       macro avg      0.666     0.585     0.589      2484
    weighted avg      0.668     0.592     0.597      2484
```

Fig. 8. Gaussian Noise Robustness Classification Results

CLASSIFICATION REPORT – Salt & Pepper Noise

ResNet50 (S&P)

	precision	recall	f1-score	support
accuracy			0.827	2484
macro avg	0.866	0.826	0.833	2484
weighted avg	0.860	0.827	0.830	2484

EfficientNetB0 (S&P)

	precision	recall	f1-score	support
accuracy			0.959	2484
macro avg	0.960	0.959	0.959	2484
weighted avg	0.961	0.959	0.959	2484

Fig. 9. SP Noise Robustness Classification Results

V. Conclusions and Future Work

A. Conclusion

Overall, under our current experimental setup, EfficientNetB0 performs slightly better than ResNet50 on the fine-grained dog breed classification task. On clean images, both models achieve good results, but their performance drops once noise is introduced. In particular, with the parameter settings we used, Gaussian noise causes a much larger decrease in performance than salt-and-pepper (S&P) noise, and EfficientNetB0 still shows a clearer advantage under noisy conditions.

At the same time, we also noticed an issue during the experiments: the two types of noise are not really comparable in terms of strength. In our setup, Gaussian noise is applied to all pixels with a relatively large standard deviation, while S&P noise only affects a small portion of the pixels. Because of this, the difference we observe between the two noise types should not be simply attributed to the noise type itself. A more reasonable explanation is that, given our current parameter choices, Gaussian noise creates a much more difficult evaluation scenario. This also means that the apparent robustness gap between the models, especially under S&P noise, may largely come from the experimental design rather than only from differences in model architecture. From this perspective, the results also reflect how the experimental setup can influence the conclusions we draw.

B. Future Work

Based on the conclusions above, we believe it would be important to design a more natural and better controlled experimental setup. For example, the parameters of different noise types could be adjusted to make their overall strength more comparable, so that the effect of noise type can be separated more clearly from the effect of noise intensity. It would also be useful to test on multiple datasets and repeat the experiments several times, in order to check whether the same observations remain stable.

In addition, more direct approaches could be explored, such as applying simple denoising methods before classification, or introducing noise-related strategies during training. At the same time, comparing a wider range of model architectures may help us better understand how model design is related to robustness.

Another direction is to move towards settings that are closer to real-world scenarios. For instance, one could consider mixed noise, different levels of corruption, or even situations where the noise distribution at test time differs from that during training. These extensions could help provide a more complete picture of how the models behave beyond the current experimental setup.

Overall, one key takeaway from this project is that, in robustness evaluation, the way corruption is defined can have a significant impact on the final conclusions. Therefore, in future work, more attention should be paid to the design of the experimental setup, so that the results are less influenced by specific parameter choices.

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